Computer Science & IT

Theory of Computation

Regular Languages

Lecture No. 6

By- Ravindra Sir

Recap of Previous Lecture

Topic — Regular languages

Topic

Topic

Topics to be Covered

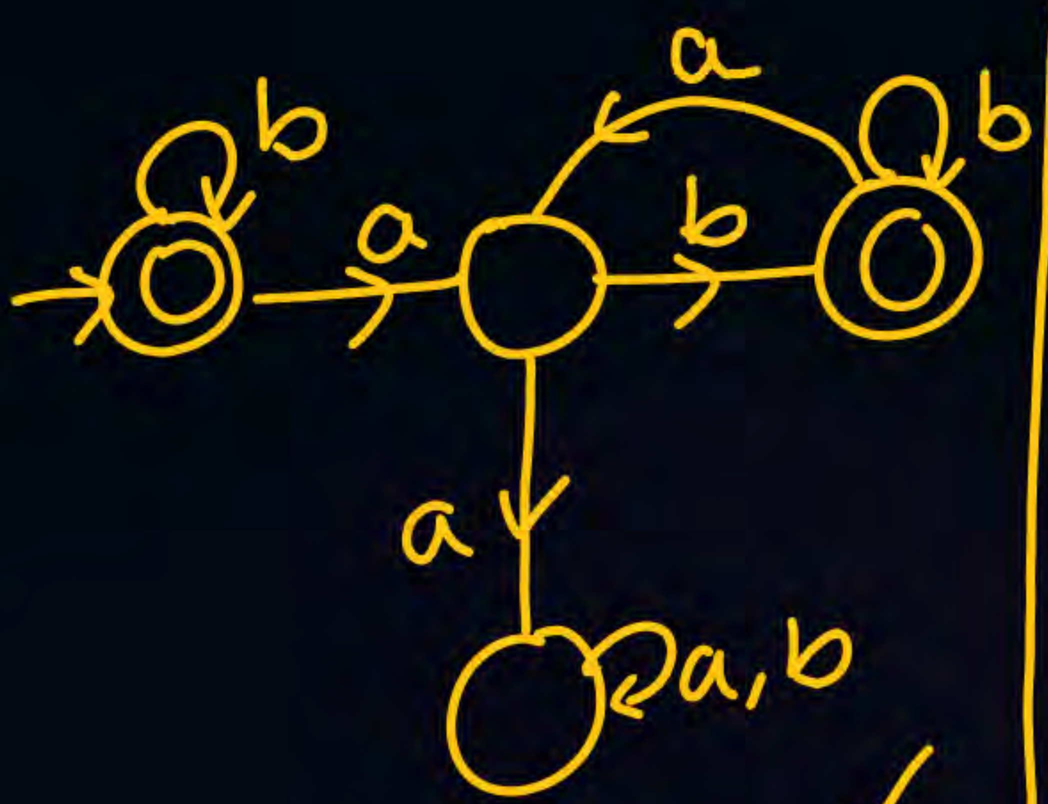
Topic

Regular languages

Topic

Topic

a is always followed by atleast 1 'b'



min DFA?

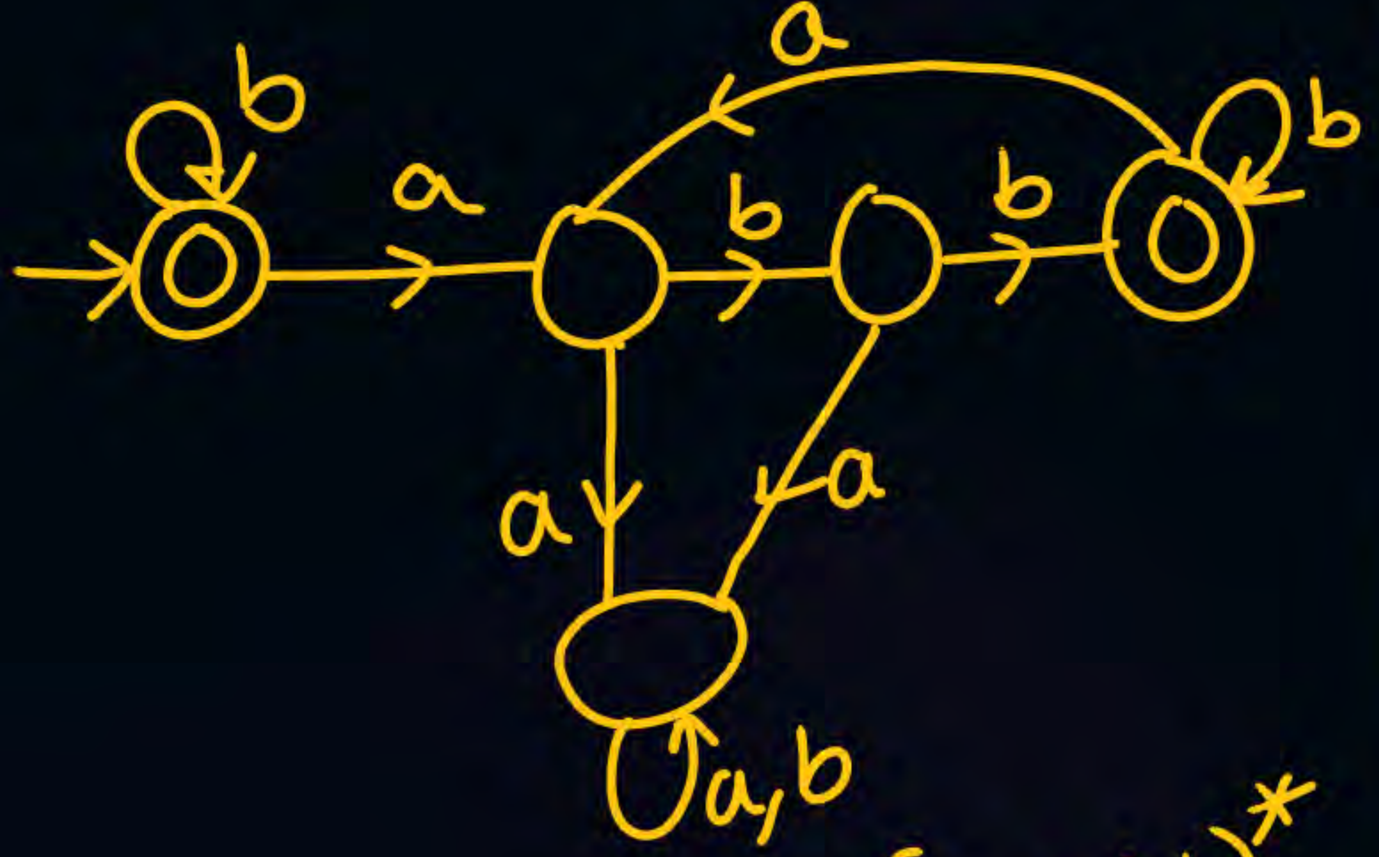
min DFA

min NFA

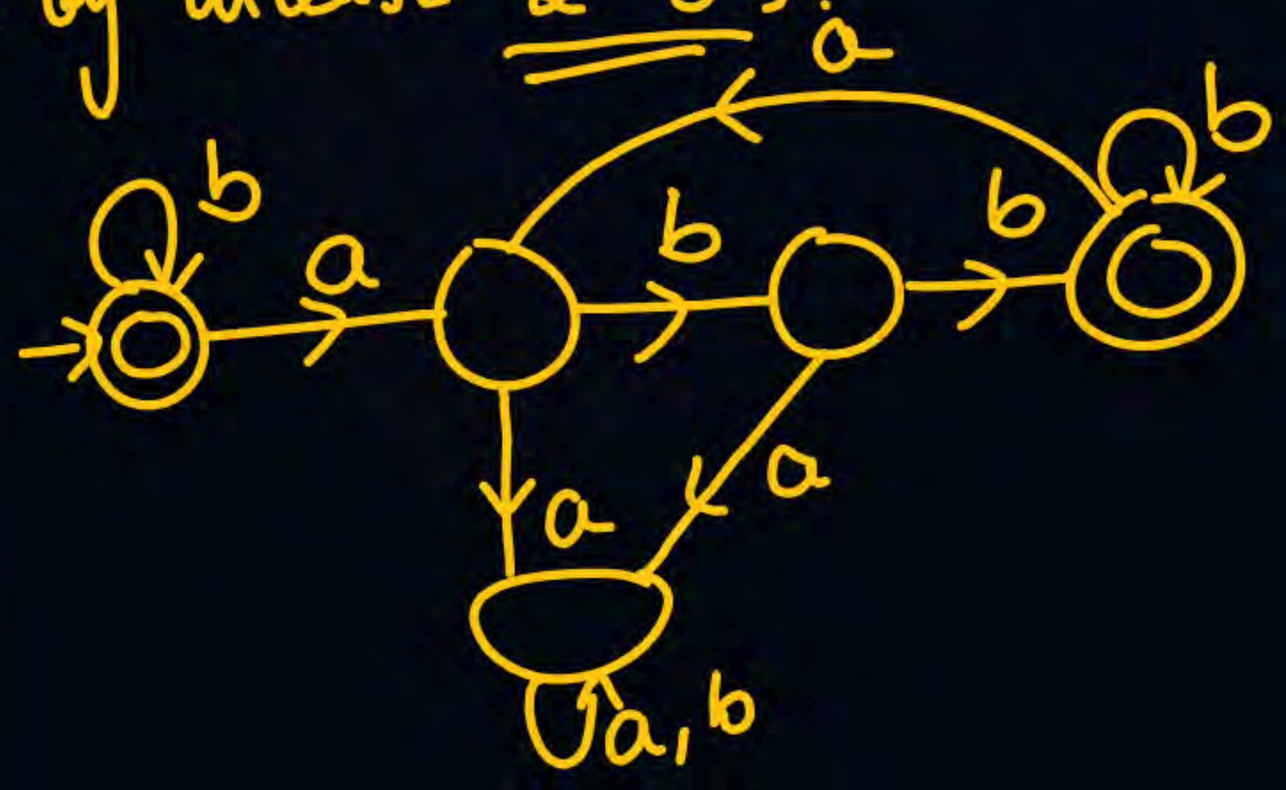
$(b + ab)^*$ aab

min of DFA

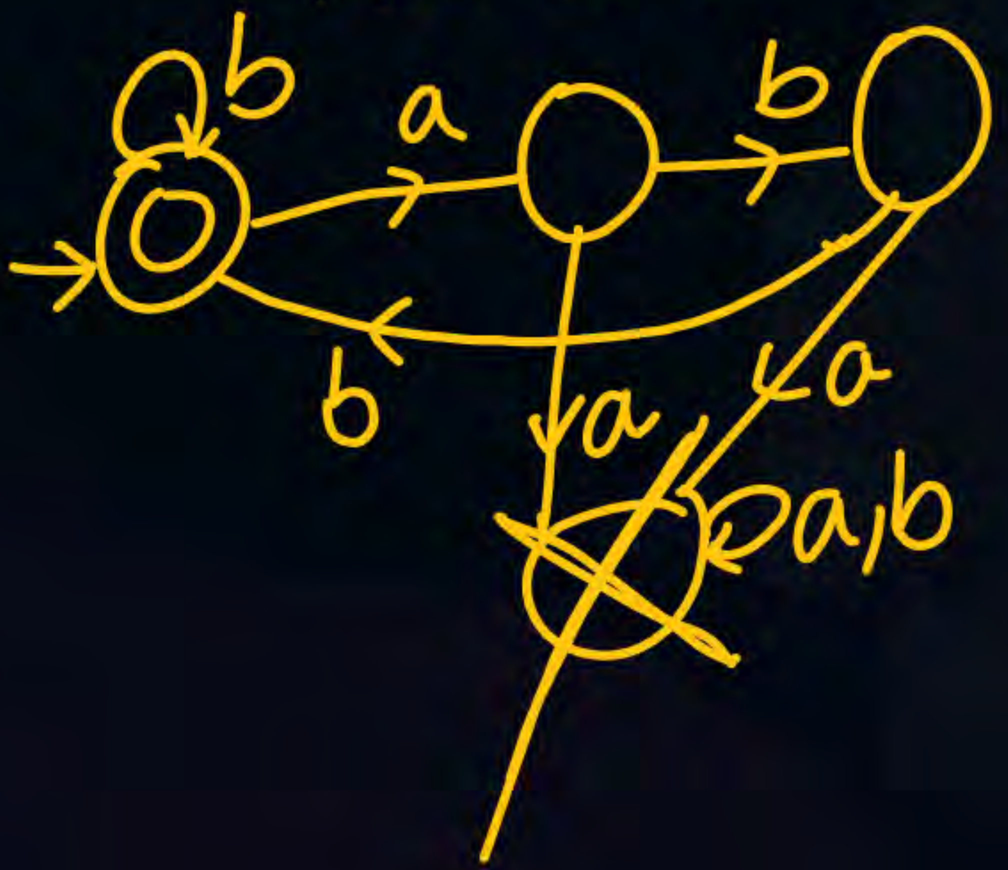
a is followed by atleast 2 b's



$(b+abb)^*$



min DFA



min DFA

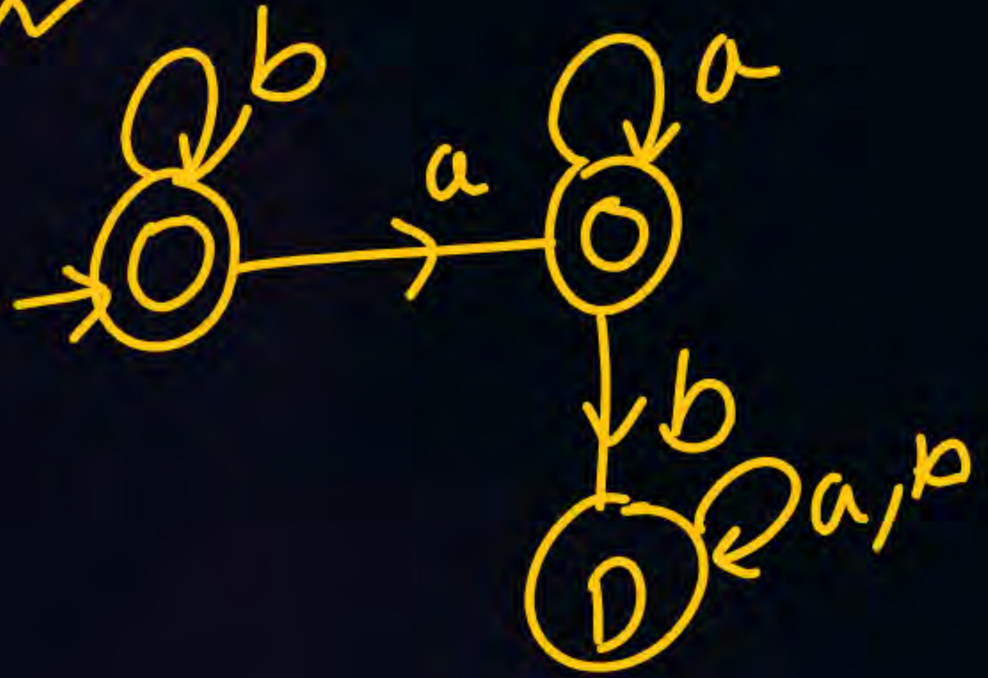
a should never be followed by a 'b'

ϵ, b, bb, bbb, ..., a, aa, aaa, ba,

$b^* a^*$ ✓

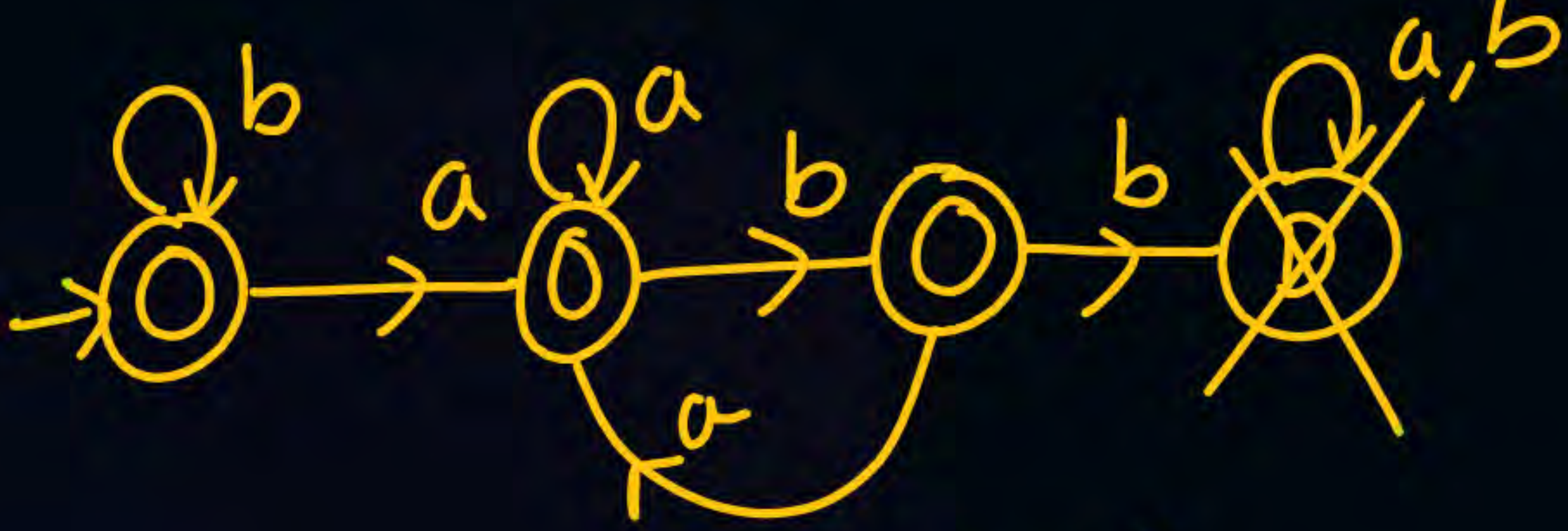
$b^* a a^*$ ✓

b^*



a is never followed by 'bb'

RE → Complex. X



min DFA → 4

min NFA → 3

$a^n b^m \,/\, n, m \geq 1$ — 3

$n, m \geq 0$ — 3

$n \geq 0, m \geq 1$ — 2

$n \geq 1, m \geq 0$ — 3

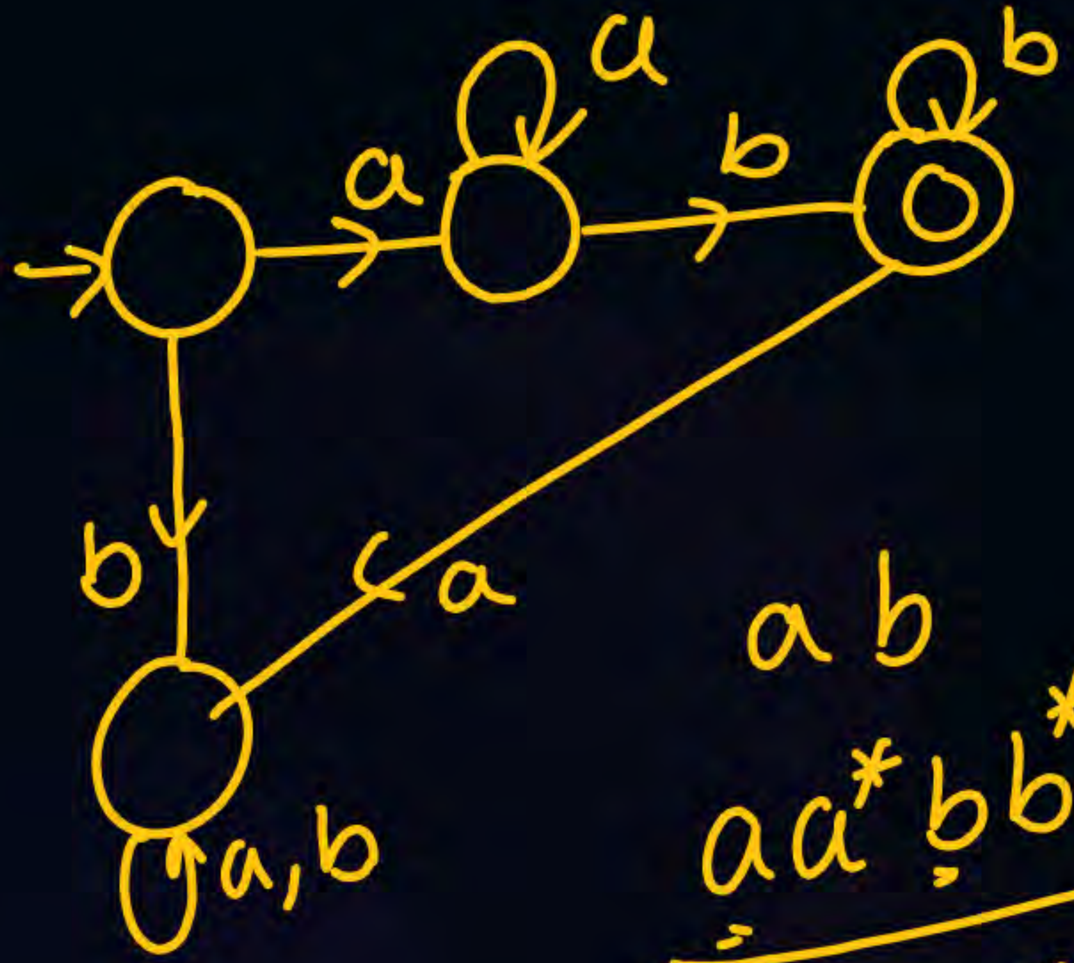
$a^n b^m c^l \,/\, n, m, l \geq 1$ — 3

$a^n b^m c^l \,/\, n, m, l \geq 0$ — 3

min DFA

→ State

15 min



ab

aa^*bb^*

$a^n b^m / n,m \geq 1$

min NFA

$\epsilon, a^*, b^*, a^*b^*$

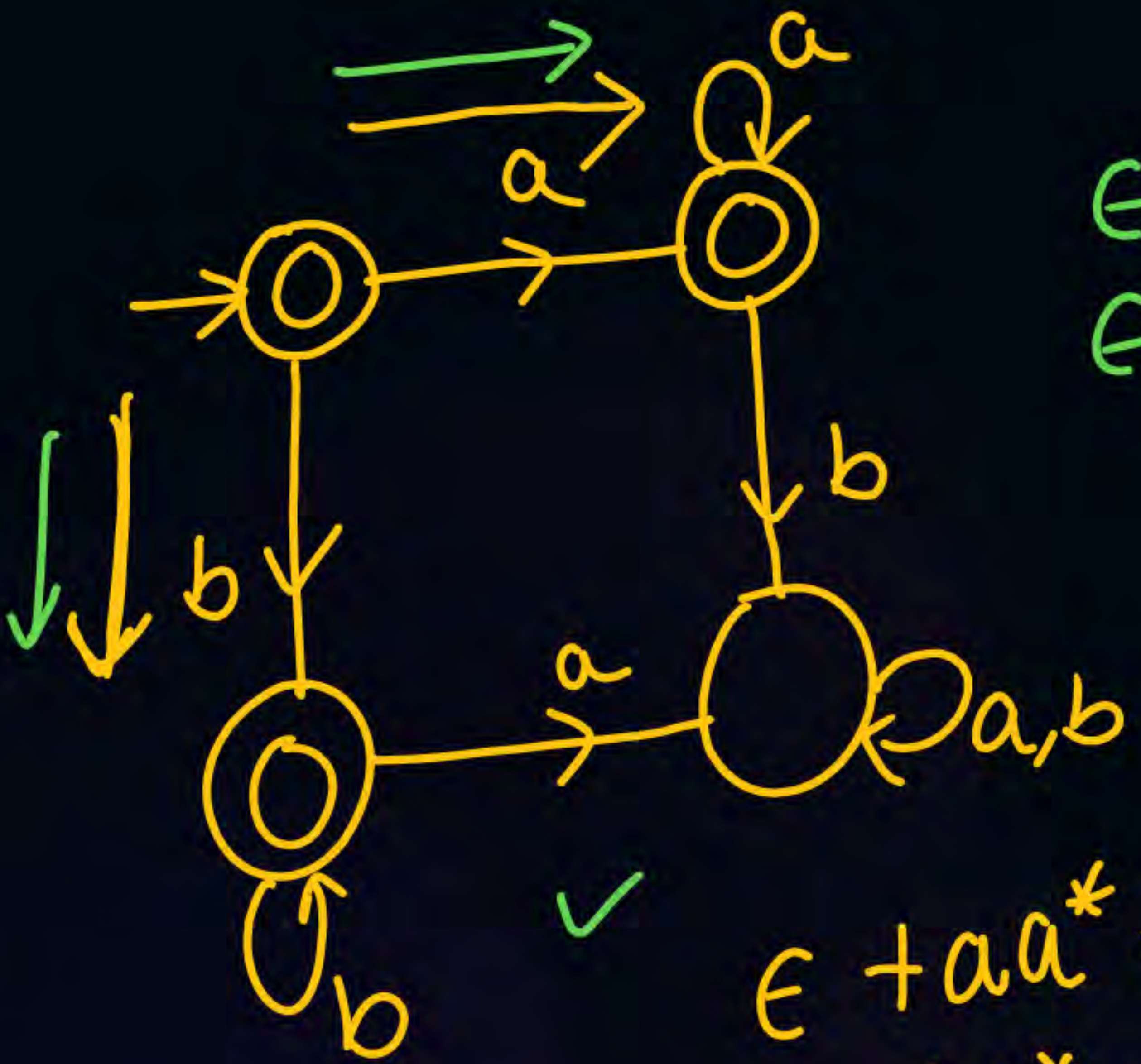
$a^* b b^* \to b^+$

$\epsilon, a^*, bb^* + \epsilon$

b^*

a^*b^*

$a^n b^m / n, m \geq 0$



$\epsilon + aa^* = a^*$ ✓

$\epsilon + bb^* = b^*$

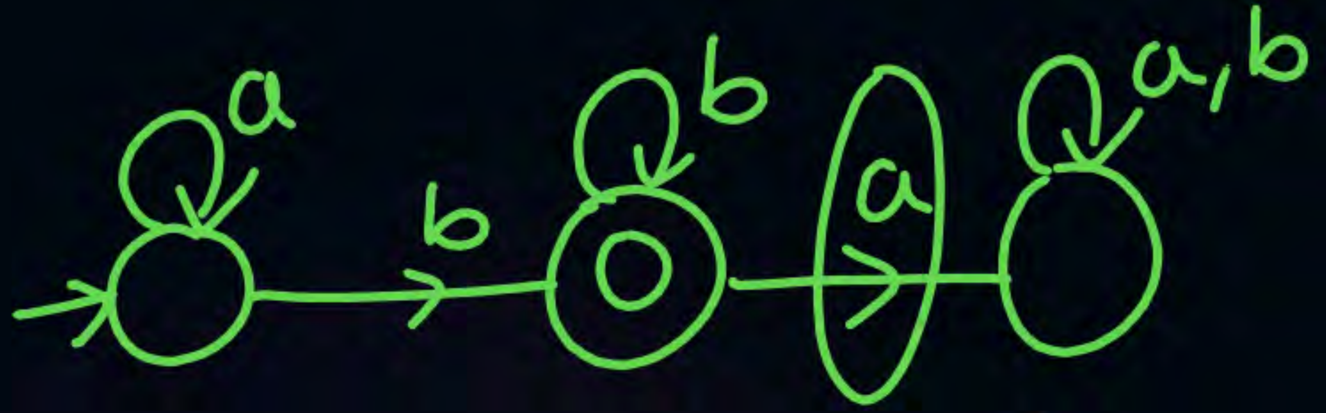
$a^* + b^*$

✓

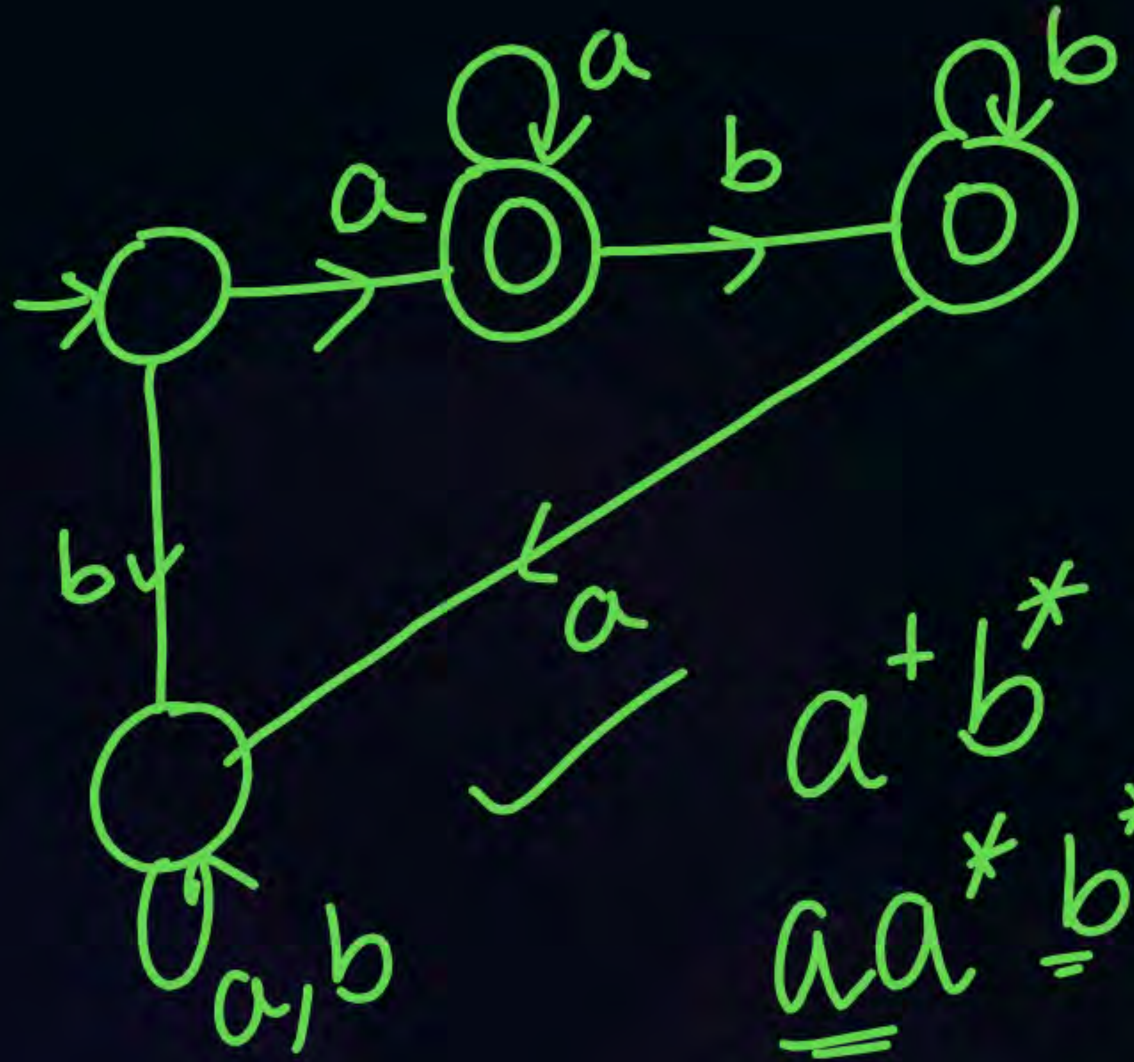
$\epsilon + aa^* = a^*$

$\epsilon + bb^* = b^*$ ✓

$a^* + b^*$



a^*bb^* $\quad$ a^*b^+



a^+b^*

aa^*b^*

aa^*bb^*

ab

a

aa^*, aa^*b^*

THANK - YOU